

a. T is a Bernoulli random variable. Hence,

$$E_p T = P_p(T = 1) = P_p\left(\sum_{i=1}^n X_i > X_{n+1}\right) = h(p).$$

b. $\sum_{i=1}^{n+1} X_i$ is a complete sufficient statistic for θ , so $E\left(T \mid \sum_{i=1}^{n+1} X_i\right)$ is the best unbiased estimator of $h(p)$. We have

$$\begin{aligned} E\left(T \mid \sum_{i=1}^{n+1} X_i = y\right) &= P\left(\sum_{i=1}^n X_i > X_{n+1} \mid \sum_{i=1}^{n+1} X_i = y\right) \\ &= P\left(\sum_{i=1}^n X_i > X_{n+1}, \sum_{i=1}^{n+1} X_i = y\right) / P\left(\sum_{i=1}^{n+1} X_i = y\right). \end{aligned}$$

The denominator equals $\binom{n+1}{y} p^y (1-p)^{n+1-y}$. If $y = 0$ the numerator is

$$P\left(\sum_{i=1}^n X_i > X_{n+1}, \sum_{i=1}^{n+1} X_i = 0\right) = 0.$$

If $y > 0$ the numerator is

$$P\left(\sum_{i=1}^n X_i > X_{n+1}, \sum_{i=1}^{n+1} X_i = y, X_{n+1} = 0\right) + P\left(\sum_{i=1}^n X_i > X_{n+1}, \sum_{i=1}^{n+1} X_i = y, X_{n+1} = 1\right)$$

which equals

$$P\left(\sum_{i=1}^n X_i > 0, \sum_{i=1}^n X_i = y\right) P(X_{n+1} = 0) + P\left(\sum_{i=1}^n X_i > 1, \sum_{i=1}^n X_i = y-1\right) P(X_{n+1} = 1).$$

For all $y > 0$,

$$P\left(\sum_{i=1}^n X_i > 0, \sum_{i=1}^n X_i = y\right) = P\left(\sum_{i=1}^n X_i = y\right) = \binom{n}{y} p^y (1-p)^{n-y}.$$

If $y = 1$ or 2 , then

$$P\left(\sum_{i=1}^n X_i > 1, \sum_{i=1}^n X_i = y-1\right) = 0.$$

And if $y > 2$, then

$$P\left(\sum_{i=1}^n X_i > 1, \sum_{i=1}^n X_i = y-1\right) = P\left(\sum_{i=1}^n X_i = y-1\right) = \binom{n}{y-1} p^{y-1} (1-p)^{n-y+1}.$$

Therefore, the UMVUE is

$$E \left(T \left| \sum_{i=1}^{n+1} X_i = y \right. \right) = \begin{cases} 0 & \text{if } y = 0 \\ \frac{\binom{n}{y} p^y (1-p)^{n-y} (1-p)}{\binom{n+1}{y} p^y (1-p)^{n-y+1}} = \frac{\binom{n}{y}}{\binom{n+1}{y}} = \frac{1}{(n+1)(n+1-y)} & \text{if } y = 1 \text{ or } 2 \\ \frac{(\binom{n}{y} + \binom{n}{y-1}) p^y (1-p)^{n-y+1}}{\binom{n+1}{y} p^y (1-p)^{n-y+1}} = \frac{\binom{n}{y} + \binom{n}{y-1}}{\binom{n+1}{y}} = 1 & \text{if } y > 2. \end{cases}$$